

RAG Techniques Benchmark: Retrieval Quality, Latency, and Hallucination Rate on a Shared arXiv Gold Set

A comparison of BM25, dense, hybrid (RRF), and LLM-reranked retrieval against no-retrieval and live web-search-agent baselines, measured on one fixed, chunk-labelled question set.

CORPUS	GOLD SET
20 arXiv papers · 84 chunks (abstracts only, 2025–2026) · topic: RAG methods & evaluation	8 answerable (chunk-seeded, synthetic) + 3 deliberately unanswerable
MODELS	RUN DATE / REPORT DATE
gemini-3.5-flash-lite & gemini-3.1-flash-lite (generation) · gemini-embedding-001, 3072-dim (retrieval)	2026-09-04 / 2026-09-05

SUMMARY. Dense retrieval alone reaches the best measured recall ($\text{Recall}@1 = \text{MRR} = 1.00$) at a steady-state cost of ~480 ms/query. Fusing it with BM25 via Reciprocal Rank Fusion regresses recall to BM25's own level (0.88) at the same latency, and LLM reranking recovers that loss at roughly 60× the latency of plain dense. Retrieval-augmented answers score 15–18 points higher on keyword recall than a no-retrieval baseline and always attach a citation, but asserted a supported claim on all 3 deliberately unanswerable questions, versus the baseline's correct abstention on 2 of 3. A parallel arm of live web-search agents (different model, full-paper access rather than cached abstracts) is included for reference, not as a controlled comparison.

1 Retrieval technique — quality and latency

Recall@K and MRR are computed against the 8 answerable gold questions, each labelled with its exact source chunk. Latency times `retriever.retrieve()` only (no claim extraction or verification) across all 11 gold questions (8 answerable + 3 unanswerable); $k=5$, with hybrid+rerank pooling 15 candidates before an LLM listwise rerank.

Table 1. Retrieval quality, $n = 8$ answerable questions.

MODE	RECALL@1	RECALL@3	RECALL@5	MRR
bm25	0.88	0.88	0.88	0.88
dense	1.00	1.00	1.00	1.00
hybrid (RRF, $k=60$)	0.88	0.88	0.88	0.88
hybrid + rerank	1.00	1.00	1.00	1.00

Table 2. Retrieval latency (ms unless noted), n = 11 queries. Index build is one-time per notebook version, not per query.

MODE	P50	P95	MEAN	MAX	INDEX BUILD
bm25	0.23	0.26	0.21	0.26	1 ms
dense	481	54,613 [†]	5,407 [†]	54,613 [†]	3,066 ms (84 chunks, 1 batched call)
hybrid	481	492	479	492	reuses dense index
hybrid + rerank	30,080	38,998	28,956	38,998	reuses dense index

[†] Inflated by client-side rate-limit queueing: the 84-chunk corpus embed and the 11 query embeds shared one 90-request/min free-tier quota window. Steady-state per-query cost is the p50 (~480 ms).

Reading Tables 1–2 together: hybrid RRF and dense cost the same per query (~480 ms) but hybrid scores like bm25 (0.88) — fusing in the weaker sparse ranking erases dense's advantage for free. Hybrid+rerank recovers dense's recall but at ~60× the latency, for one additional LLM call per query. Dense alone dominates on this corpus: top recall at the cheaper tier.

Table 3. Claim-level citation-verifier calibration, n = 8 answerable questions.

MODE	CITATION PRECISION	CITATION COMPLETENESS	UNSUPPORTED RATE
bm25	1.00	1.00	0.00
hybrid	0.94	1.00	0.00
hybrid + rerank	0.90	1.00	0.05

Unsupported rate stays ~0 across every mode: the verifier downgrades disputable claims to partial rather than rejecting them, so as a filter it currently blocks almost nothing.

2 Direct vs. RAG vs. web-search agents

Same 8 answerable + 3 unanswerable questions, graded at the answer level. `naive_llm` and `hybrid_rag(+rerank)` ran on `gemini-3.1-flash-lite` after `gemini-3.5-flash-lite` returned sustained 503 UNAVAILABLE mid-run. `web_search_agents` is four parallel Claude subagents (WebSearch/WebFetch, budget-capped at 4 calls of each per question), scored with the identical `keyword_recall` / `expresses_uncertainty` functions used above — but is not a controlled comparison (see §3, limitation 2).

Table 4. Answer-quality by condition, n = 8 answerable + 3 unanswerable.

CONDITION	KEYWORD RECALL	CITATION AVAILABILITY	HALLUCINATION RATE
naive_llm (no retrieval)	0.62	0.00	0.33
hybrid_rag	0.77	1.00	1.00
hybrid_rag + rerank	0.80	1.00	1.00
web_search_agents	1.00	1.00	0.00

Reading Table 4: RAG improves keyword recall by 15–18 points over the no-retrieval baseline and always attaches a citation, but asserted a supported claim on all 3 unanswerable questions — worse than the no-retrieval baseline, which correctly hedged on 2 of 3. Retrieval reliably finds *something* topically nearby; nothing downstream is currently allowed to say “not in this corpus.”

Table 5. Search-agent arm, per question

Q	TYPE	KW. RECALL	OUTCOME
Q1– Q8	answerable	1.00 (each)	all keywords hit, cited
Q9	unanswerable	—	abstained
Q10	unanswerable	—	abstained
Q11	unanswerable	—	abstained — false premise found [*]

^{*} Q11 asked for per-model F1 scores from a paper that reports none; agents verified this by reading the full text rather than guessing.

Headline investigation (qualitative, n=3 claims)

A single free-text run of the pipeline's own question (“what RAG methods reduce hallucination, and how is RAG quality evaluated?”) on hybrid+rerank produced 3 claims: 1 supported, 2 partial — 33% citation precision. Sample too small to trend; included for completeness, not comparison.

3 Limitations

- Abstract-only ingestion is the structural ceiling.** Every arXiv source here is chunked from its abstract alone (no PDF/HTML ingestion). Citation verification can support “X was proposed,” never “X scored Y on benchmark Z,” because Results sections are never ingested.
- The web-search-agent row is not a like-for-like arm.** It differs from the RAG arms in both model (Claude agents vs. Gemini) and corpus depth (live full papers vs. cached abstracts), so it isolates abstention behavior rather than demonstrating “search beats RAG.” Its keyword_recall saturated at 1.00 on every answerable question — the metric ran out of headroom, not every answer was equally good.
- n is small.** 8 answerable / 3 unanswerable questions over one 20-paper corpus. Gold questions are chunk-seeded (generated from the same chunk they are labelled against), which is exact but likely easier and more lexically similar to the source than a real user question — retrieval numbers across the board should be read as an upper bound.
- Dense p95/max latency (Table 2) is a rate-limit artifact, not a property of the embeddings API.** It appears only because corpus embedding and query embedding shared one free-tier quota window in this run; a warm, already-embedded notebook pays the p50.
- call_with_backoff did not retry 5xx before this run.** A sustained 503 UNAVAILABLE from gemini-3.5-flash-lite twice killed the citation/baseline sections; the retry path now covers 5xx as well as 429, and the model was switched to gemini-3.1-flash-lite to complete this run — so Table 4's RAG/naive rows and Table 3 are not on the same model as Table 1's headline investigation.

4 Reproduction

Retrieval quality (Tables 1, 3) + answer quality (Table 4, RAG/naive rows)	<code>run_topic_eval.py</code>	<code>rag-methods</code>
Retrieval latency (Table 2)	<code>run_latency_bench.py</code>	<code>rag-methods</code>
Search-agent arm (Table 4/5, <code>web_search_agents</code> row)	<code>score_search_agents.py</code>	<code>rag-methods</code> <code>answers.json</code>
Raw logs: <code>results/2026-09-04_rag-methods_*</code> . Full architecture and prior-run notes: <code>STACK.md</code> .		